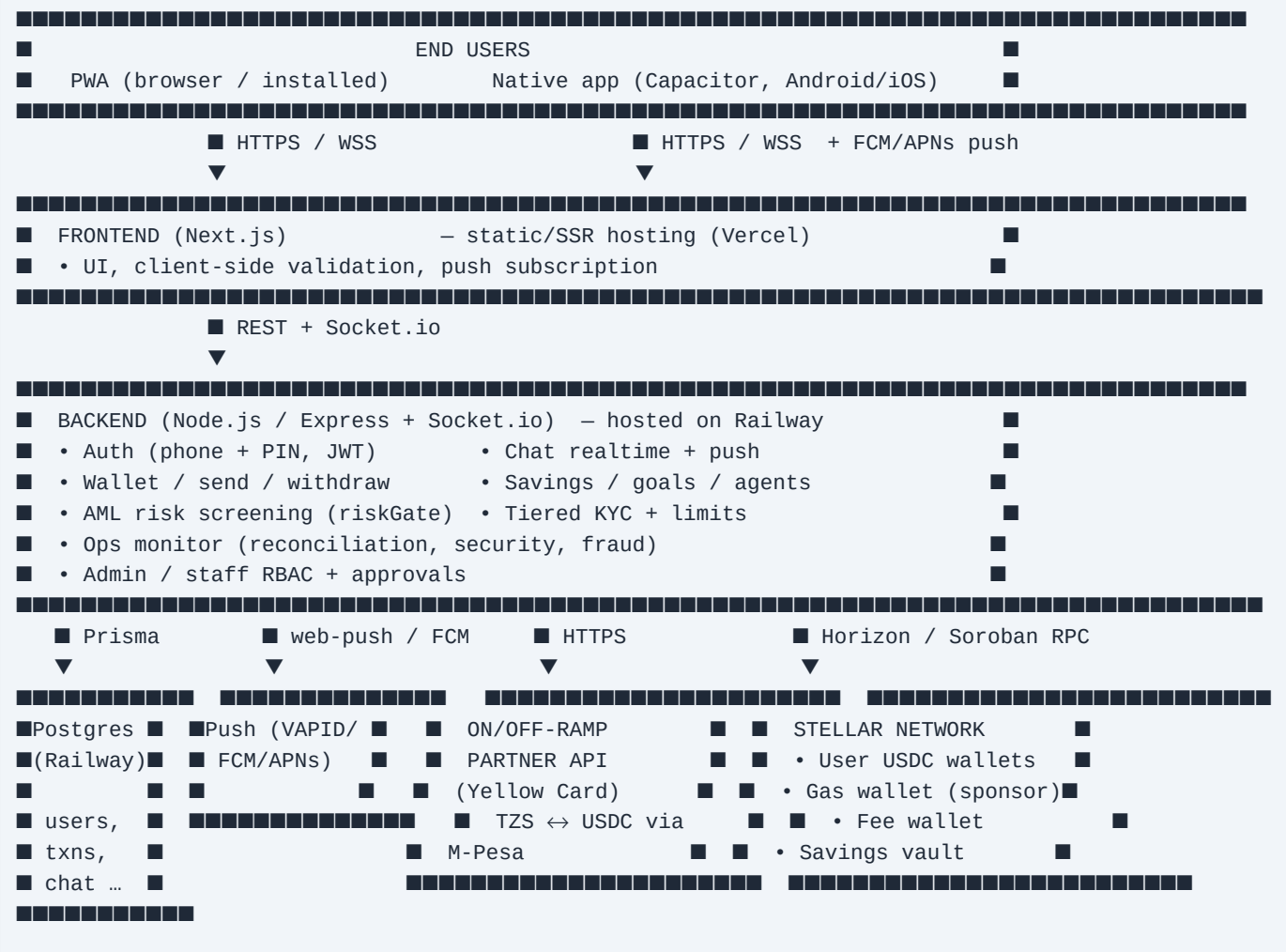


02 — Technical Architecture

1. High-level overview

OlomiPay is a three-tier system: a client (web PWA + native mobile shell), a backend API + realtime service, and a settlement layer on the Stellar network. The local fiat (M-Pesa) leg is handled entirely by the **licensed on/off-ramp partner**; OlomiPay never directly touches the mobile-money rails.



2. Components & technology

Layer	Technology	Hosting
Web frontend	Next.js 14 (App Router), React, TypeScript	Vercel
Native apps	Capacitor (Android/iOS) loading the live web app	App stores / APK
Backend API	Node.js, Express, TypeScript	Railway

Realtime	Socket.io (WebSocket + long-polling, optional Redis adapter)	Railway
Database	PostgreSQL via Prisma ORM	Railway (managed)
Settlement	Stellar (USDC asset); Soroban for token transfers	Public Stellar network
Push	Web Push (VAPID) + FCM (Android) / APNs (iOS)	Google/Apple
Error/uptime	Sentry; internal ops monitor + alerting	—

3. Settlement layer (Stellar)

- Users hold **USDC** in individual Stellar accounts.
- A **gas wallet** sponsors account reserves and pays network fees via fee-bump transactions, so users never need to hold the native asset (XLM).
- A **dedicated fee wallet** (separate from the gas wallet) collects the platform fee with its own USDC trustline.
- A **savings vault** address holds funds placed into savings/goals.
- Outbound user→user transfers are signed with the user's key (PIN-authorized); platform-originated movements (e.g. savings withdrawal payout) are signed by platform-controlled wallets. See Doc 04 for the key model.

4. Authentication & sessions

- **End users:** phone number + 6-digit PIN → JWT access token (+ long-lived refresh token). PINs are stored only as **bcrypt** hashes.
- **Staff/admin:** separate username + password identities with **role-based access control** (super-admin, department heads, department staff) and a **multi-approver workflow** for sensitive actions.
- Account lockout after repeated failed attempts (see Doc 05/06).

5. Integrations

Integration	Purpose	Direction
On/off-ramp partner (Yellow Card)	TZS ↔ USDC via M-Pesa (deposit/withdraw/remit)	OlomiPay → partner API + partner webhooks
Circle (optional/treasury)	USD ↔ USDC treasury; bank payouts	OlomiPay → Circle API
Stellar Horizon / Soroban RPC	On-chain balances, transfers	OlomiPay → network

FCM / APNs / Web Push	Notifications	OlomiPay → device
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6. Environments & deployment

- Source control: Git (GitHub). CI/CD via Vercel (frontend) and Railway (backend).
- Secrets are held as platform environment variables (see Doc 06 for handling).
- Separate testnet/sandbox configuration is supported for partner sandbox testing.

7. Data stored (high level)

- User: phone, name, KYC level/details, encrypted wallet key, Stellar public key.
- Transactions: type, amount (USDC), status, counterparties, timestamps, refs.
- Chat: conversations, messages (see Doc 06 re content handling), receipts.
- Compliance: security events, risk reviews, admin audit log, approvals.
- Full schema available to the partner on request under NDA.